

Convolutional neural networks

ImageNet (image-net.org)

Soccer, association football

A football game in which two teams of 11 players try to kick or head a ball into the opponents' goal

1395
pictures

90.76%
Popularity
Percentile



Numbers in brackets: (the number of synsets in the subtree).

- ImageNet 2011 Fall Release (32326)
 - plant, flora, plant life (4486)
 - geological formation, formation (1)
 - natural object (1112)
 - sport, athletics (176)
 - rowing, row (2)
 - funambulism, tightrope walking (0)
 - judo (0)
 - blood sport (10)
 - gymnastics, gymnastic exercise (0)
 - water sport, aquatics (19)
 - track and field (5)
 - outdoor sport, field sport (17)
 - contact sport (18)
 - boxing, pugilism, fisticuffs (0)
 - wrestling, rassling, grappling (0)
 - ice hockey, hockey, hockey (0)
 - football, football game (5)
 - rugby, rugby football, rug (0)
 - professional football (0)
 - American football, Amer (0)
 - soccer, association foot (0)
 - team sport (0)
 - racing (7)
 - athletic game (70)
 - riding, horseback riding, equit (0)
 - archery (0)
 - cycling (3)
 - sledding (3)
 - skating (6)

Treemap Visualization

Images of the Synset

Downloads

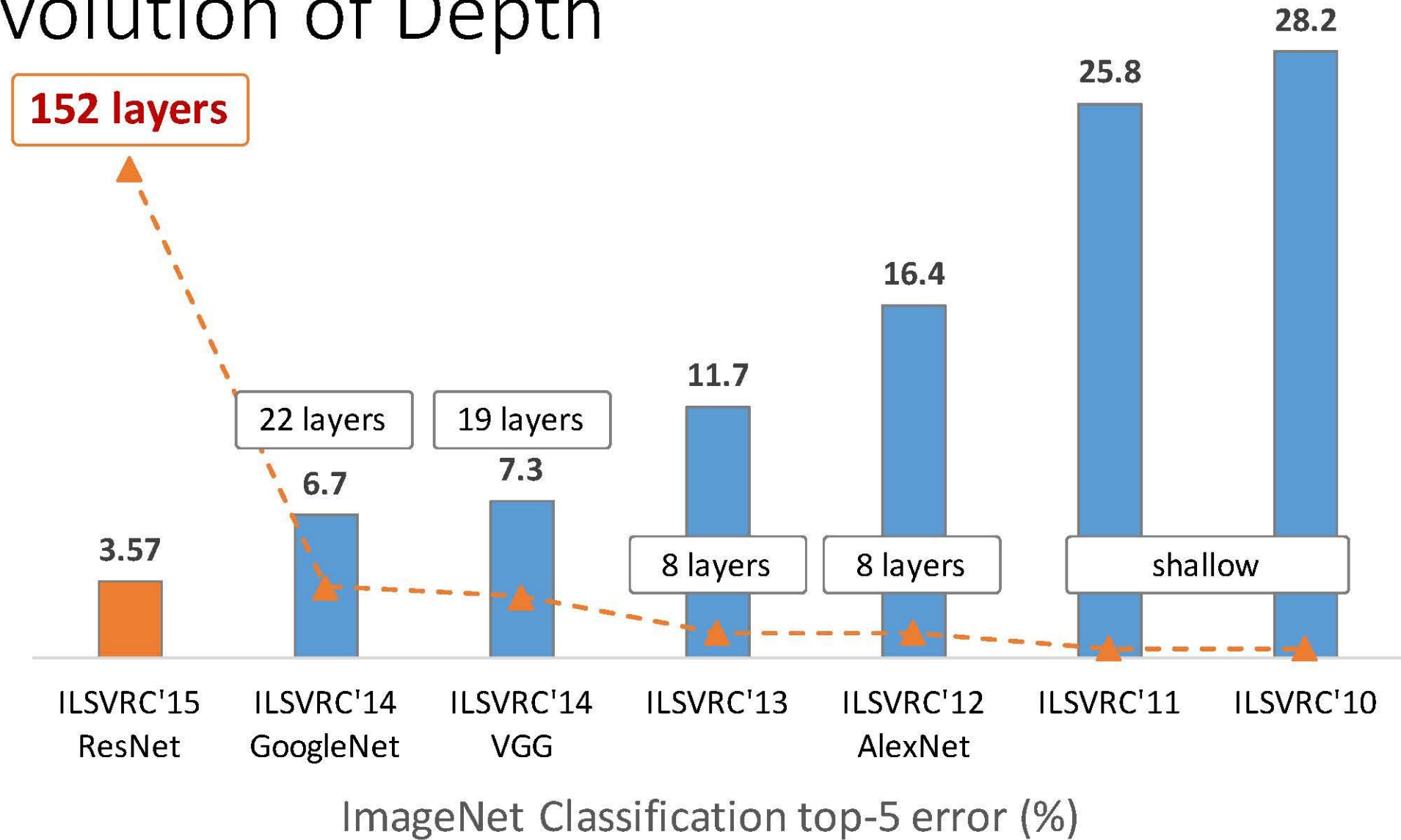


*Images of children synsets are not included. All images shown are thumbnails. Images may be subject to copyright.

Prev 1 2 3 4 5 6 7 8 9 10 ... 39 40 Next

14,197,122 images
21841 categories

Revolution of Depth



Sobel filter

Initial image

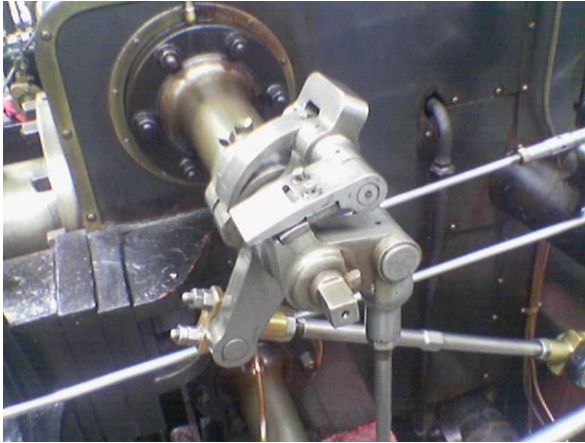
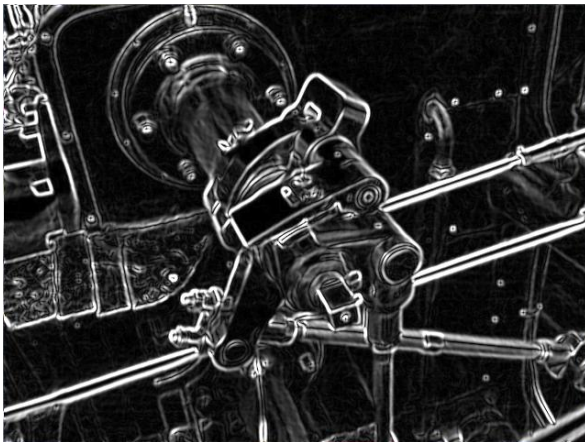


Image after the filter application



1. Compute horizontal brightness gradients

$$I_x = I * G_x \quad G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}$$

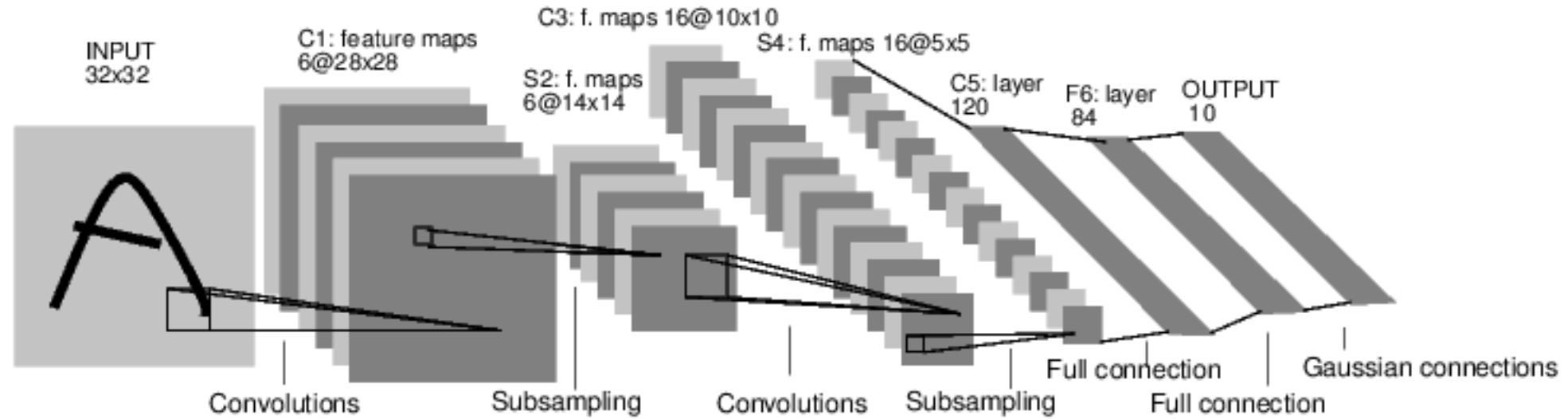
2. Compute vertical brightness gradients

$$I_y = I * G_y \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix}$$

3. Final image

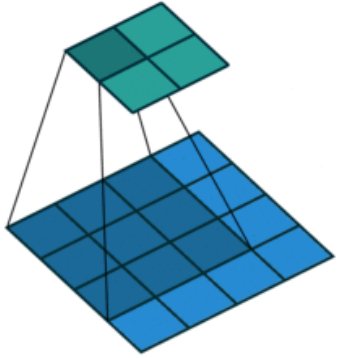
$$G = \sqrt{I_x^2 + I_y^2}$$

LeNet network [LeCun et al., 1998]

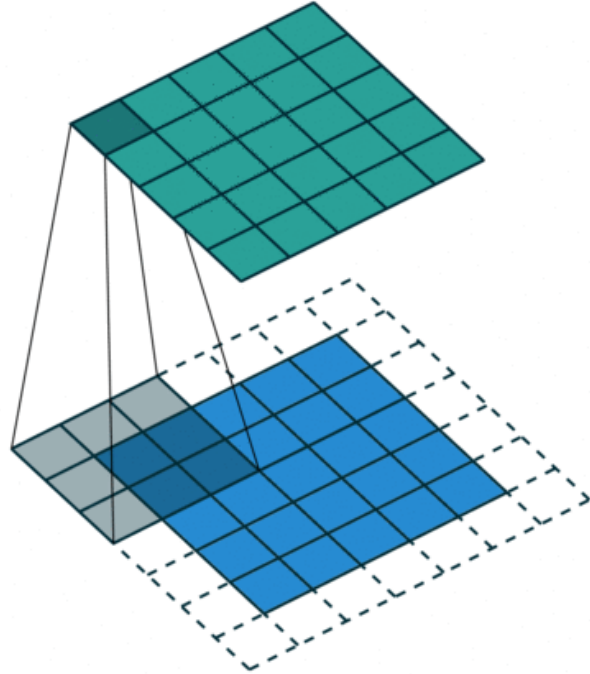


- Operations in the network:
 - Convolution with kernel
 - Non-linearity
 - Pooling
 - Fully-connected networks

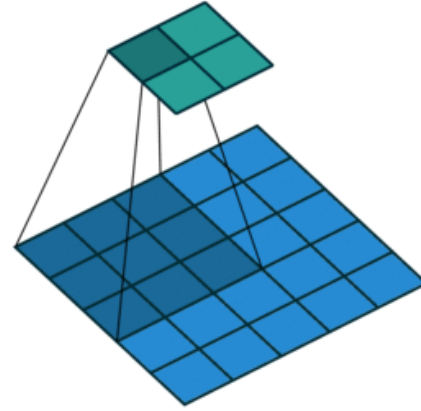
Convolution visualization



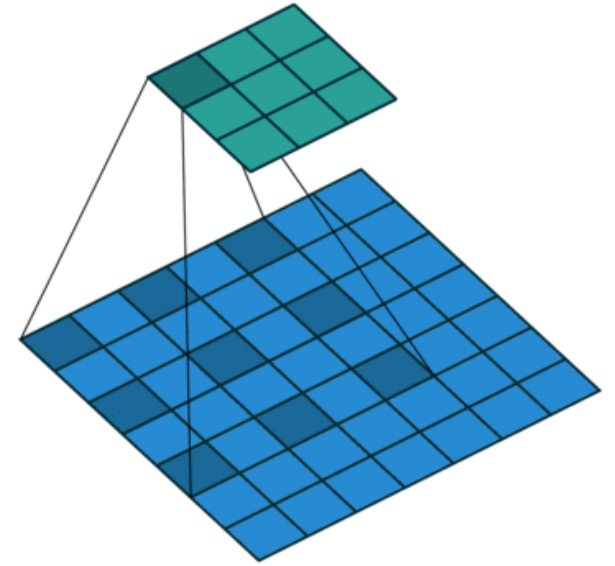
No padding, no stride



Zero padding, no stride

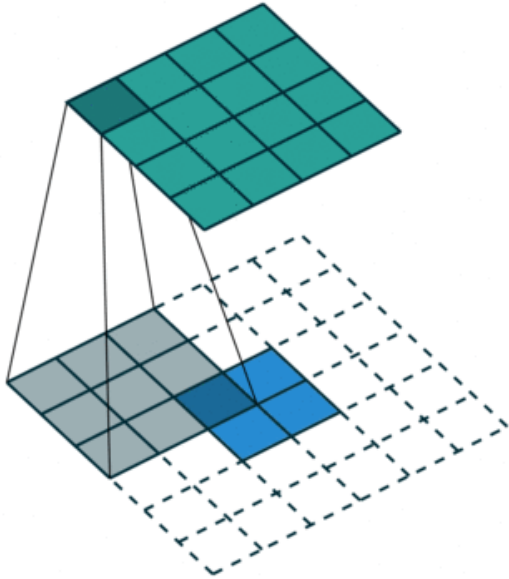


No padding,
stride = 2

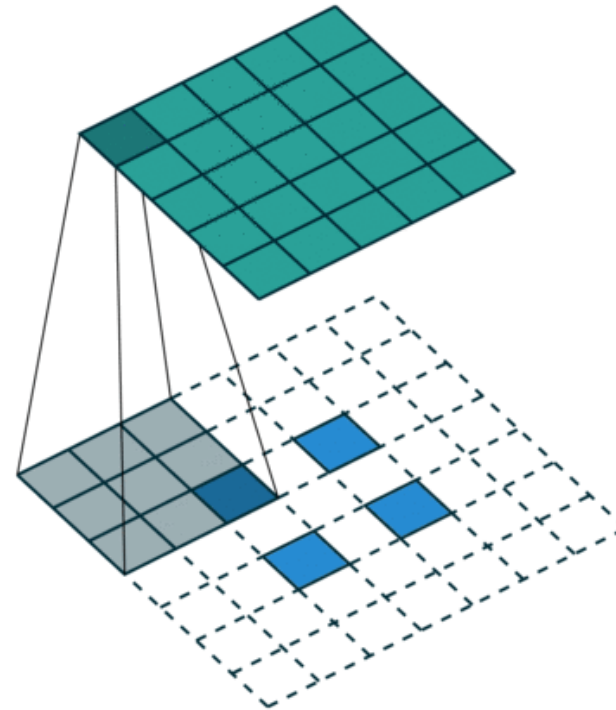


No padding, no stride,
dilation = 2

Convolution visualization

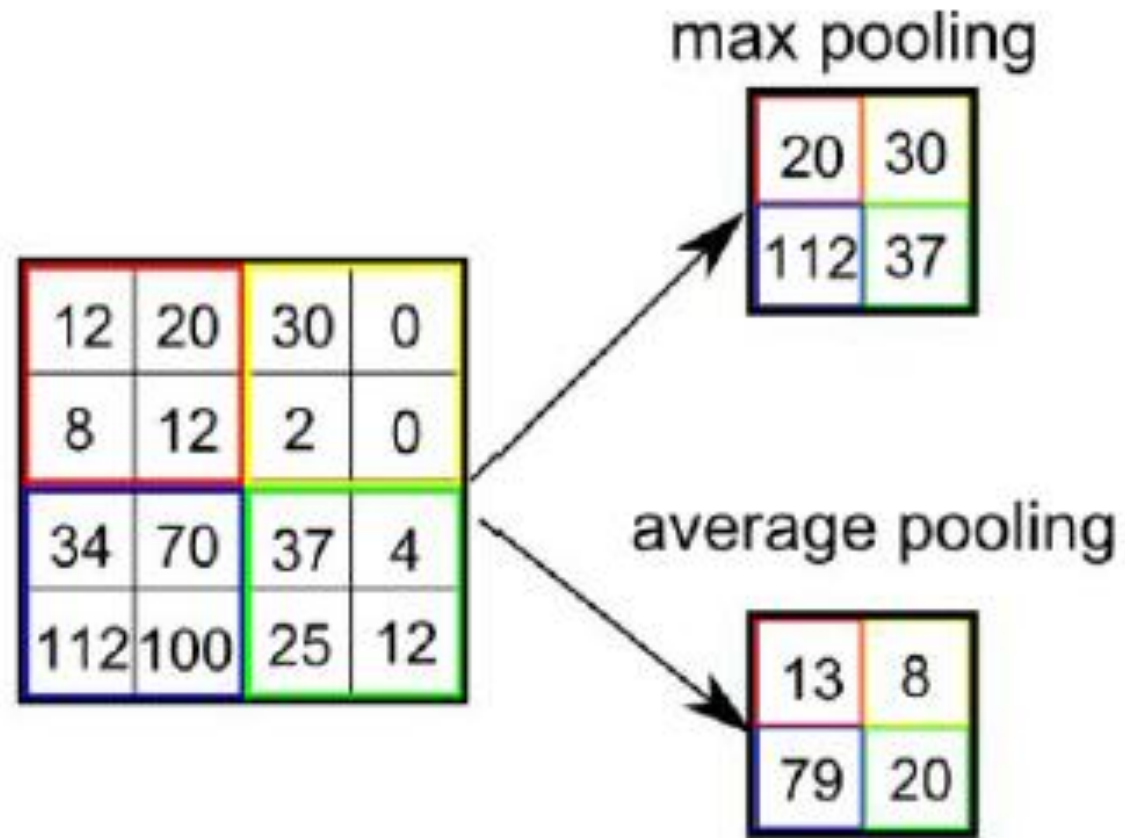


Convolution transposed
No padding, no stride

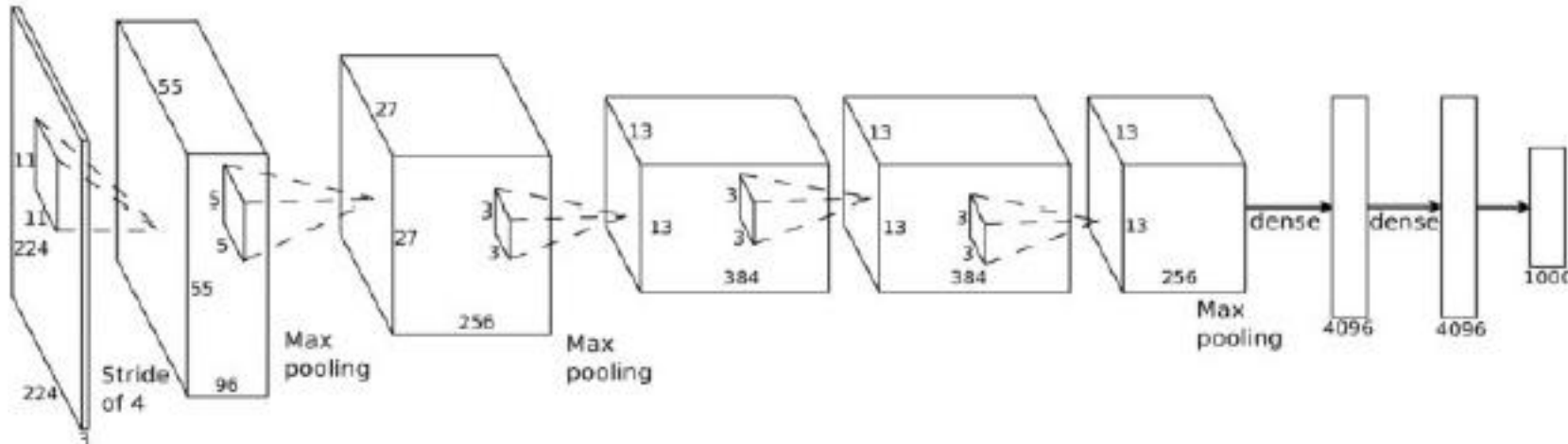


Convolution transposed
No padding, stride = 2

Pooling operation



AlexNet network (2012)

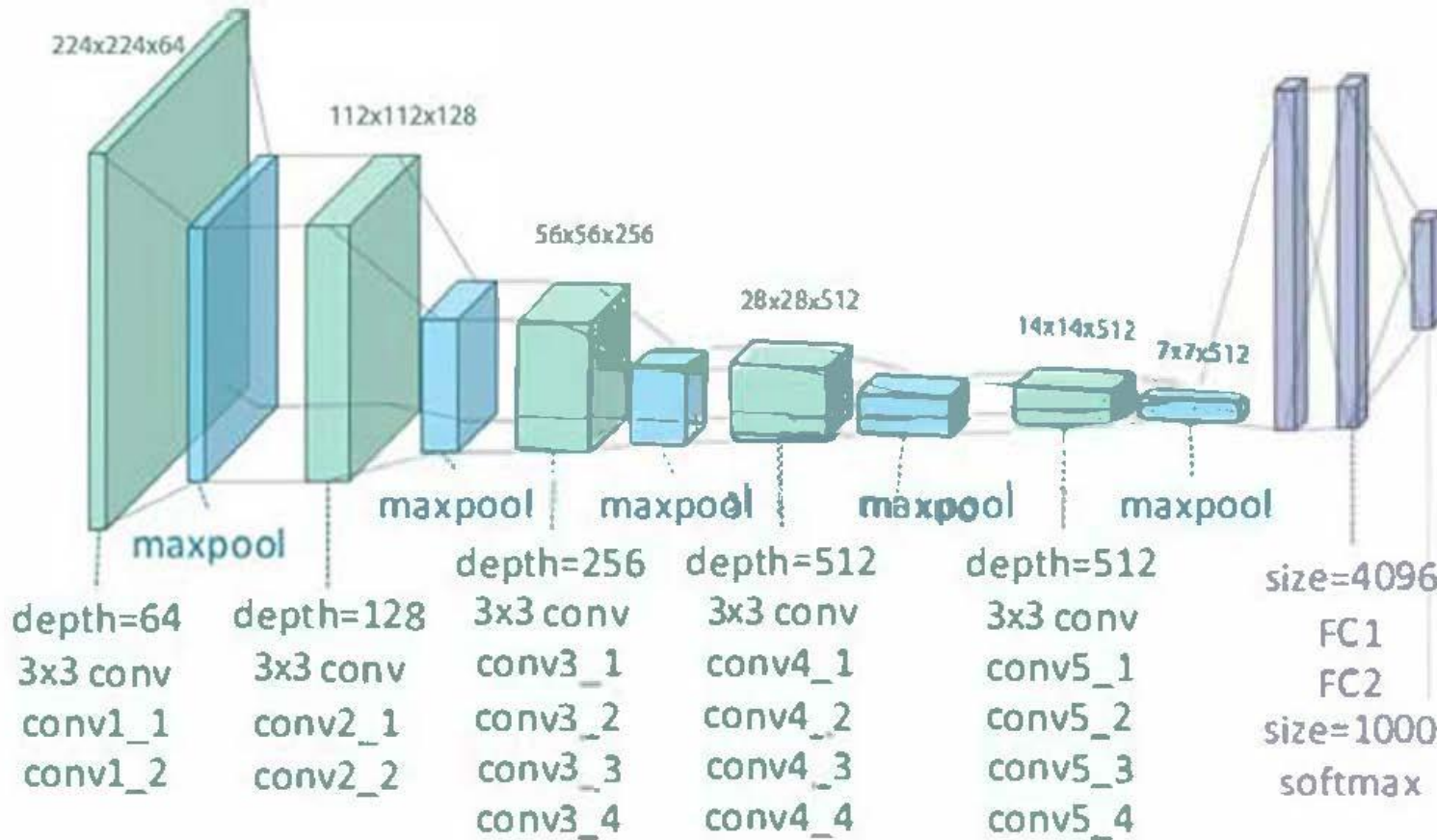


- Max pooling
- ReLu nonlinearity
- Fully connected layers
- More data and bigger model (60M params)
- + Data augmentation (flips and random samples)
- + Dropout regularization
- + GPUs (50x speed up)
- + 1 week of training on 2 GPUs

Data augmentation



VGG network (2014)



- Cascade of 3x3 convs instead of big convs
- 140M parameters (AlexNet has 60M)
- More balanced computations
- Can't be trained from scratch (gradient vanishes)
- Several training stages for different depths
- Training on 4 Titan Black GPUs for 2-3 weeks

ResNet network (2015)

AlexNet, 8 layers
(ILSVRC 2012)



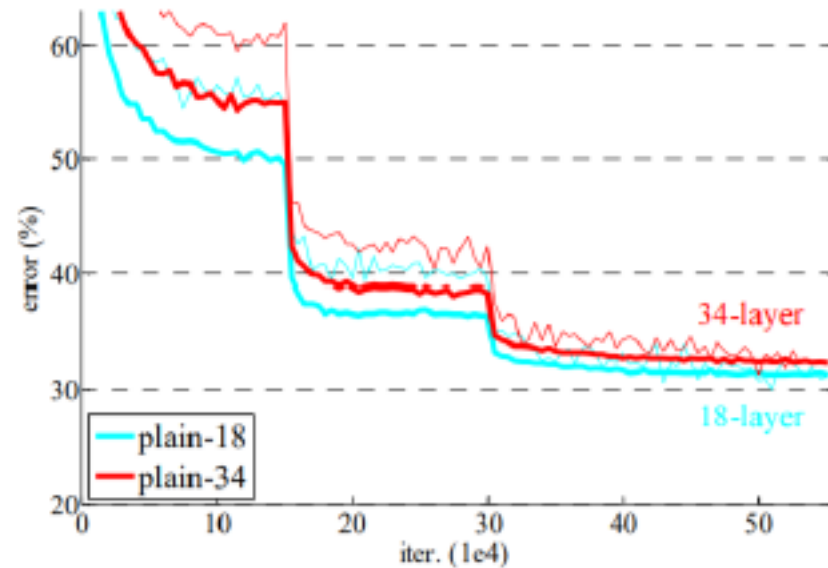
ResNet, **152 layers**
(ILSVRC 2015)

VGG, 19 layers
(ILSVRC 2014)



ResNet network (2015)

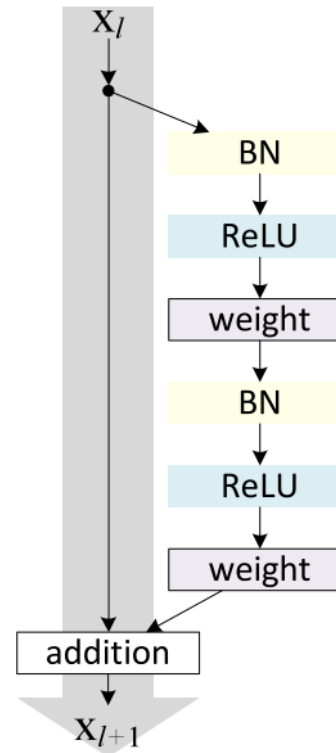
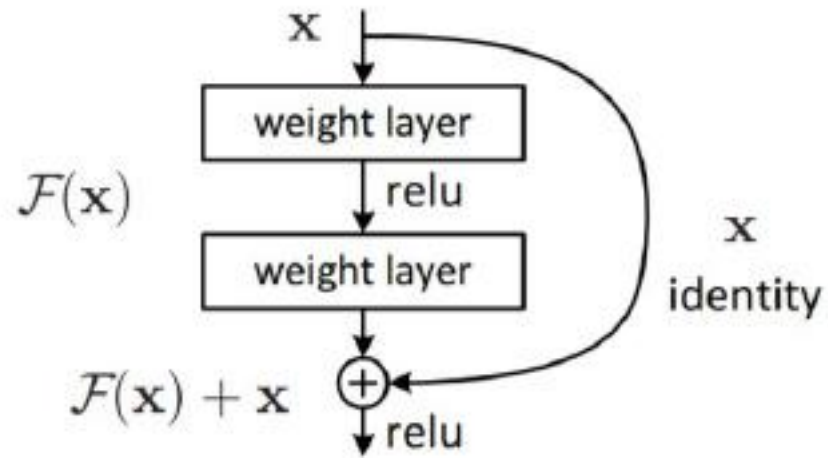
Just adding layers doesn't work



ResNet, **152 layers**
(ILSVRC 2015)

ResNet network (2015)

- The main idea – residual block = adding identity connection
- Allows better propagate gradients inside



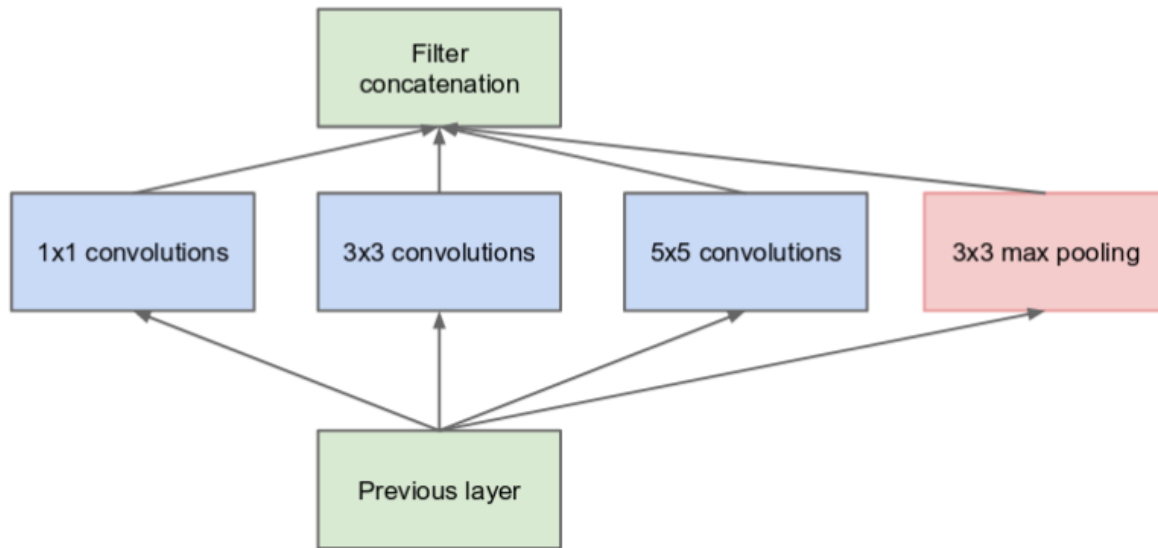
ResNet, **152 layers**
(ILSVRC 2015)

ResNet network (2015)

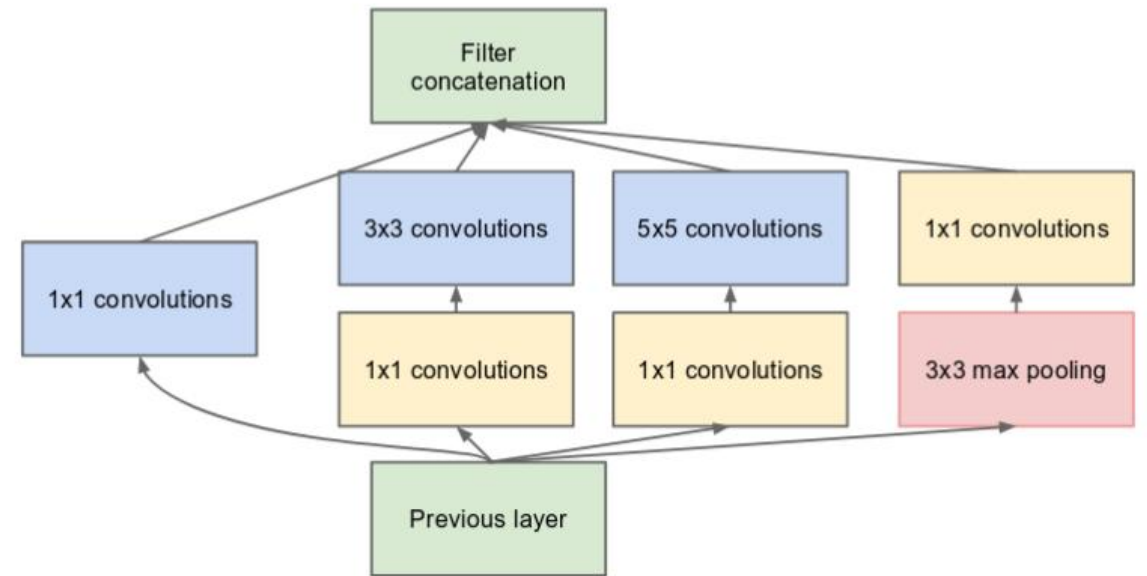
layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
conv2_x	56×56	3×3 max pool, stride 2				
		$\begin{bmatrix} 3\times 3, 64 \\ 3\times 3, 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times 3, 64 \\ 3\times 3, 64 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 64 \\ 3\times 3, 64 \\ 1\times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 64 \\ 3\times 3, 64 \\ 1\times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 64 \\ 3\times 3, 64 \\ 1\times 1, 256 \end{bmatrix} \times 3$
conv3_x	28×28	$\begin{bmatrix} 3\times 3, 128 \\ 3\times 3, 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times 3, 128 \\ 3\times 3, 128 \end{bmatrix} \times 4$	$\begin{bmatrix} 1\times 1, 128 \\ 3\times 3, 128 \\ 1\times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1\times 1, 128 \\ 3\times 3, 128 \\ 1\times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1\times 1, 128 \\ 3\times 3, 128 \\ 1\times 1, 512 \end{bmatrix} \times 8$
conv4_x	14×14	$\begin{bmatrix} 3\times 3, 256 \\ 3\times 3, 256 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times 3, 256 \\ 3\times 3, 256 \end{bmatrix} \times 6$	$\begin{bmatrix} 1\times 1, 256 \\ 3\times 3, 256 \\ 1\times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1\times 1, 256 \\ 3\times 3, 256 \\ 1\times 1, 1024 \end{bmatrix} \times 23$	$\begin{bmatrix} 1\times 1, 256 \\ 3\times 3, 256 \\ 1\times 1, 1024 \end{bmatrix} \times 36$
conv5_x	7×7	$\begin{bmatrix} 3\times 3, 512 \\ 3\times 3, 512 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times 3, 512 \\ 3\times 3, 512 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 512 \\ 3\times 3, 512 \\ 1\times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 512 \\ 3\times 3, 512 \\ 1\times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 512 \\ 3\times 3, 512 \\ 1\times 1, 2048 \end{bmatrix} \times 3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

ures for ImageNet. Building blocks are shown in brackets (see also Fig. 5), with the numbers of block

Inception network (2014-2016)

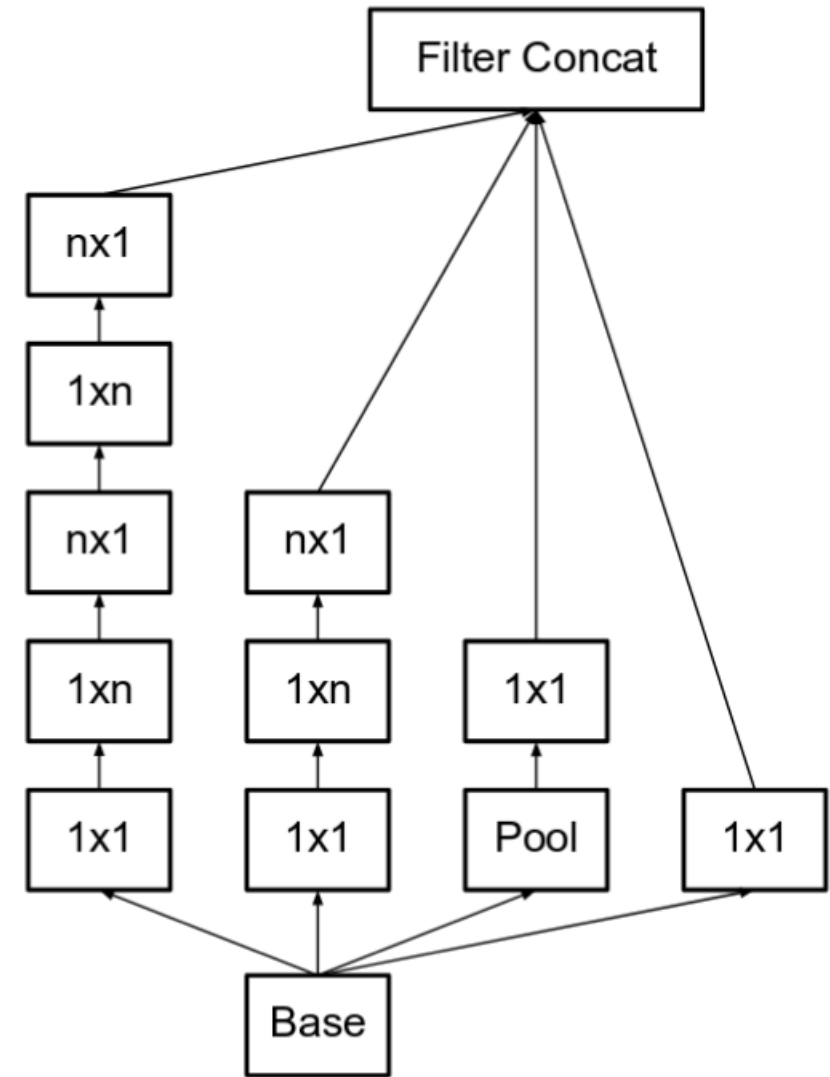
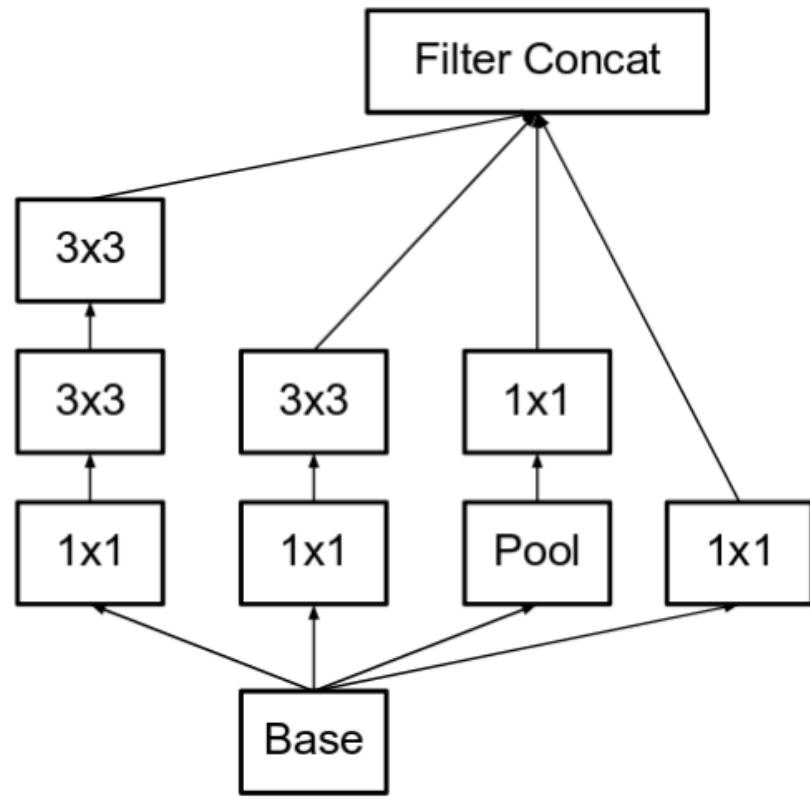


(a) Inception module, naïve version

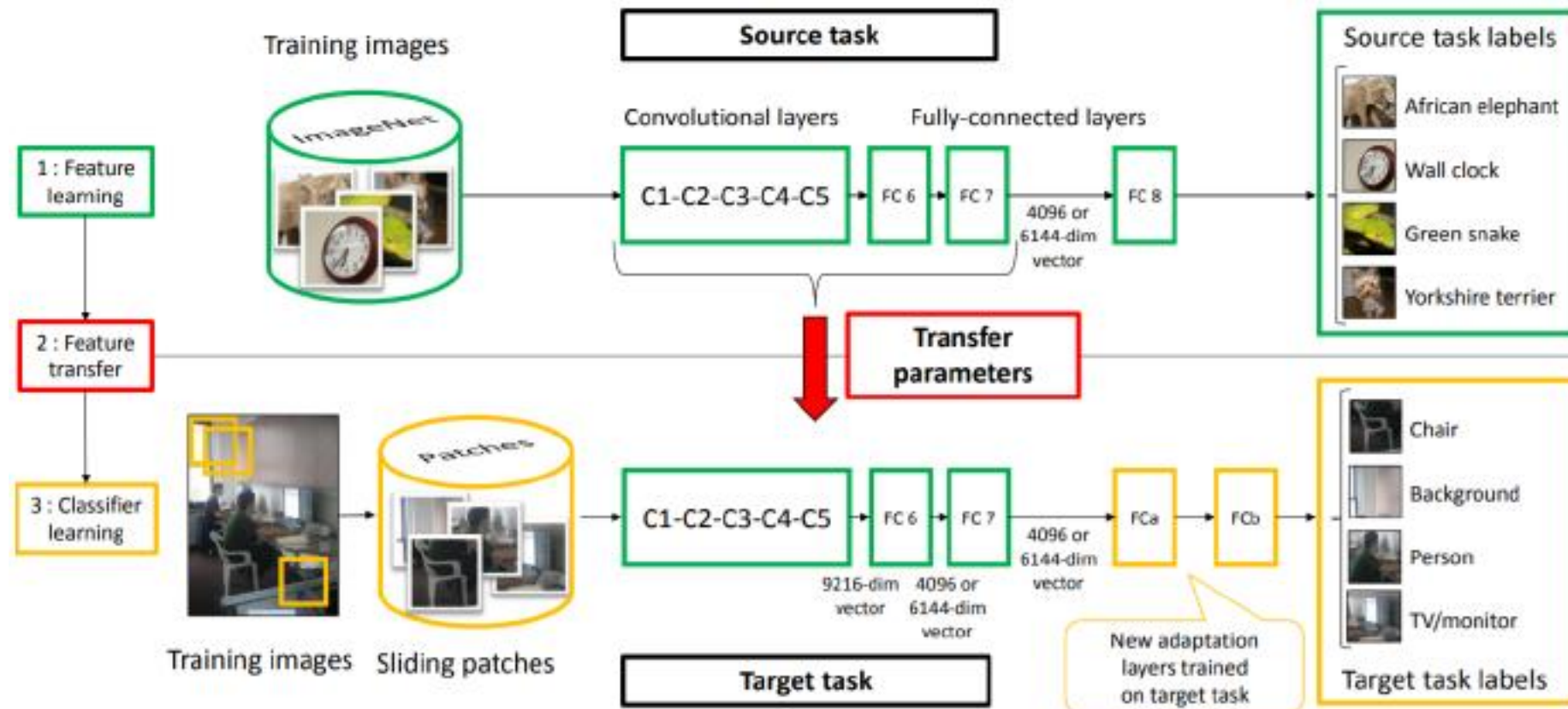


(b) Inception module with dimension reductions

Inception network (2014-2016)

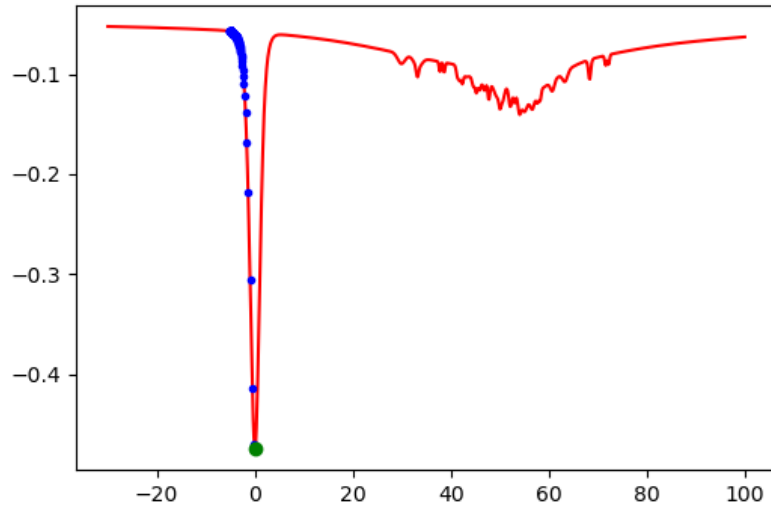


Transfer learning in CNN [Oquab et al., 2014]

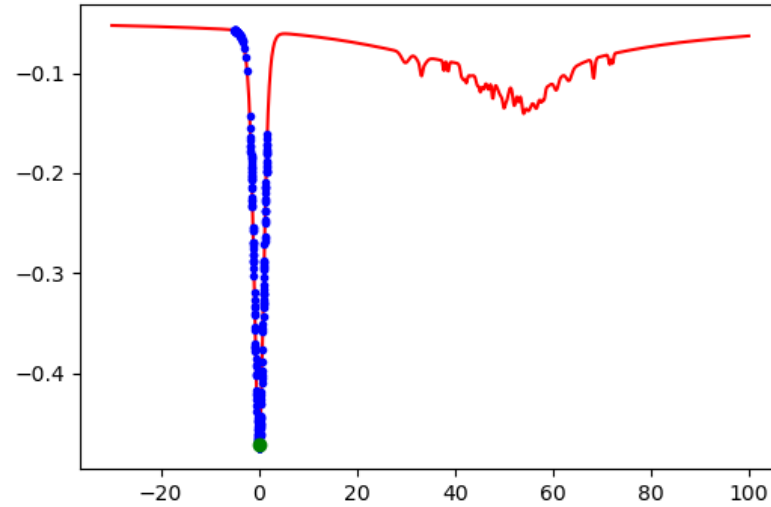


Optimization with SGD

Full GD with proper step size



SGD with little noise



SGD with larger noise

